

what models, backgrounds, objects, you create and how you control their movement. Key framing is the technique to time the movement of your scene where you define start position, end positions and attributes of the objects such as size, rotation etc.

Creating dope sheets and animation workflow

Once you decide the key poses of the character, you need to time these poses. You can arrange your poses on something like an exposure (X) sheet, also called a dope sheet. On this sheet you write the instructions about each frame. Now take your rigged model and keyframe all the poses. Once you've got these poses keyed in, take a quick preview.

Tweening

Tweening is creation of successive frames of animation between key frames. Here, intermediate frames between 2 images are generated to allow smooth transition. Flash automatically creates these intermediary frames using motion tween or shape tween options.

Onion skinning

While you edit your animation, you can see several previous frames along with the current frame. This technique is called onion skinning and the animator can adjust the animation based on previous frames.

- ▶ **Motion capture:** An actor generally wears a skintight suit that has markers on it and performs the actions. The computer can easily pick up the markers as the actor is being shot from different camera angles. The motion of the markers is being recorded as motion data. In the studio, they import this data into popular 3D software to animate the model. Special care needs to be taken to match the rig of the model and the actor for real and smooth animation.
- ▶ **Particle effect systems:** Games can contain general effects like fire or smoke that only serve as visual appeal or generally informative purpose. These effects may be running as animated sequences of 3D objects. The effects are usually created with particle effect systems that are automatically repeated throughout the whole game.

Tools used

- ▶ Adobe Flash
- ▶ Daz Studio, Poser
- ▶ 3Ds Max